

NAME

mbset – Sets values in **mbprocess** parameter files.

VERSION

Version 5.0

SYNOPSIS

mbset **-Iinfile** [**-E** **-Fformat** **-L** **-N** **-PPARAMETER:value** **-V** **-H**]

DESCRIPTION

mbset is a utility for creating and modifying **mbprocess** parameter files.

The program **mbprocess** can perform a variety of swath data processing functions in a single step (producing a single output swath data file), including:

- Merge edited navigation generated by **mbnavedit**.
- Apply bathymetry edit flags from **mbedit** and **mbclean**
- Recalculate bathymetry from raw travel time and angle data by raytracing through water sound speed models from **mbvelocitytool** or **mbsvplist**.
- Apply changes to roll bias, pitch bias, heading bias, and draft values.
- Recalculate sidescan from raw backscatter samples (Simrad multibeam data only).
- Apply tides to bathymetry.
- Insert metadata.

The actions of **mbprocess** are controlled by text parameter files. Each **mbprocess** parameter file contains single line commands that set processing modes and parameters. The **-P** option of **mbset** is used to modify a single **mbprocess** parameter command. This option can be invoked as many times as desired on the command line, allowing **mbset** to set multiple **mbprocess** processing parameters and modes. If the swath data file specified by the **-Iinfile** option of **mbset** has an existing **mbprocess** parameter file, then that parameter file will be read and the existing parameter values will be modified. If no **mbprocess** parameter file exists, then **mbset** starts with default processing parameters, modifies those, and then generates a new parameter file.

If the input file specified by the **-I** option is a single swath data file, **mbset** will act on that file and its parameter file only. If the input file is a datalist, then **mbset** will act on all swath files extracted from the datalist (and any recursively parsed datalist files). This function allows users to easily set uniform processing parameters for large numbers of data files. The default input file is "datalist.mb-1".

The processing parameter file used by **mbprocess** has an ".par" suffix. In addition to **mbset**, these files are generated or modified by **mbedit**, **mbnavedit**, **mbvelocitytool**, **mbnavadjust**, and **mbclean**.

The processed output swath files produced by **mbprocess** are named using a convention based on the data format id. **MB-System** data formats are specified using two-digit or three-digit numbers (see the **MBIO** manual page). If an input swath data file is named "root.mbXX", where XX is the format id, then the default processed output file will be "rootp.mbXX" (e.g. mydata.mb71 -> mydatap.mb71). The "p" inserted before the ".mbXX" suffix indicates the output file has been created by **mbprocess**. If the input file does not follow the *.mbXX naming convention, then the output filename will just consist of the input name with "p.mbXX" added as a suffix (e.g. mydata -> mydatap.mb71)

MBPROCESS PARAMETER FILE COMMANDS

The **mbprocess** parameter file commands modified by **mbset** are:

GENERAL PARAMETERS:

EXPLICIT

- causes mbprocess to set modes implicitly
- e.g. the SVPPFILE command will also set raytracing on even if the RAYTRACE command is not given [explicit mode commands required]

FORMAT constant

- sets format id [no default]

INFILE filename

- sets input file path [no default]

OUTFILE filename

- sets output file path [no default]

NAVIGATION MERGING:

NAVMODE boolean

- sets navigation merging [0]
- 0: navigation merge off
- 1: navigation merge on

NAVFILE filename

- sets navigation file path [no default]

NAVFORMAT constant

- sets navigation file format [9]
- see the **mbprocess** man page for documentation of the supported navigation formats.

NAVHEADING boolean

- sets heading to be merged from navigation file
- note: heading merged from navigation before heading correction applied
- 0: heading not changed
- 1: heading merged from navigation file

NAVSPEED boolean

- sets speed to be merged from navigation file
- 0: speed not changed
- 1: speed merged from navigation file

NAVDRAFT boolean

- sets draft to be merged from navigation file
- note: draft merged from navigation before draft correction applied
- 0: draft not changed
- 1: draft merged from navigation file

NAVATTITUDE boolean

- sets roll, pitch and heave to be merged from navigation file
- note: roll, pitch, and heave merged from navigation before roll bias and pitch bias corrections applied
- 0: roll, pitch, and heave not changed
- 1: roll, pitch, and heave merged from navigation file

NAVINTERP boolean

sets navigation interpolation algorithm [0]

0: linear interpolation (recommended)

1: spline interpolation

NAVTIMESHIFT constant

sets navigation time shift (seconds) [0.0]

– note: time shift added to timestamps of navigation fixes read in from NAVFILE prior to merging

NAVIGATION OFFSETS AND SHIFTS:**NAVSHIFT** boolean

sets navigation offset [0]

– note: offsets and shifts are applied to navigation values from both survey and navigation records, and are applied to navigation read in from NAVFILE prior to merging

– note: offsets and shifts are NOT applied to adjusted navigation values from NAVADJFILE

NAVOFFSETX constant

sets navigation athwartship offset (meters) [0.0]

– note: the effective navigation shift is (NAVOFFSETX – SONAROFFSETX), and the navigation is corrected by subtracting this effective shift.

– note: athwartship shift is positive to starboard.

NAVOFFSETY constant

sets navigation fore-aft offset (meters) [0.0]

– note: the effective navigation shift is (NAVOFFSETY – SONAROFFSETY), and the navigation is corrected by subtracting this effective shift.

– note: fore-aft shift is positive forward.

NAVOFFSETZ constant

sets navigation vertical offset (meters) [0.0]

– note: this value is not yet used for anything.

– note: vertical shift is positive down.

NAVSHIFTLON constant

sets navigation longitude shift (degrees) [0.0]

NAVSHIFTLAT constant

sets navigation latitude shift (degrees) [0.0]

NAVSHIFTX constant

sets navigation longitude shift (meters) [0.0]

NAVSHIFTY constant

sets navigation latitude shift (meters) [0.0]

ADJUSTED NAVIGATION MERGING:**NAVADJMODE** mode

sets navigation merging from mbnadjust [0]

– can apply to longitude and latitude only or longitude, latitude, and depth offset

- 0: adjusted navigation merge off
- 1: adjusted navigation merge on
- 2: adjusted navigation and depth offset merge on

NAVADJFILE filename

sets adjusted navigation file path

- this file supercedes navigation file for lon and lat only
- uses mbnadjust output

NAVADJINTERP boolean

sets adjusted navigation interpolation algorithm [0]

- 0: linear interpolation (recommended)
- 1: spline interpolation

ATTITUDE MERGING:

ATTITUDEMODE mode

sets attitude (roll, pitch, and heave) merging [0]

- roll, pitch, and heave merged before roll bias and pitch bias corrections applied
- attitude merging from a separate file supersedes attitude merging from a navigation file
- 0: attitude merging off
- 1: attitude merging on

ATTITUDEFILE filename

sets attitude file path

ATTITUDEFORMAT constant

sets attitude file format [1]

- attitude files can be in one of four ASCII table formats
- 1: format is <time_d roll pitch heave>
- 2: format is <yr mon day hour min sec roll pitch heave>
- 3: format is <yr jday hour min sec roll pitch heave>
- 4: format is <yr jday daymin sec roll pitch heave>
- time_d = decimal seconds since 1/1/1970
- daymin = decimal minutes start of day
- roll = positive starboard up, degrees
- pitch = positive forward up, degrees
- heave = positive up, meters

SONARDEPTH MERGING:

SONARDEPTHMODE mode

sets sonardepth merging [0]

- sonardepth merged before draft corrections applied
- sonardepth merging from a separate file supersedes draft merging from a navigation file
- 0: sonardepth merging off
- 1: sonardepth merging on

SONARDEPTHFILE filename

sets sonardepth file path

SONARDEPTHFORMAT constant

sets sonardepth file format [1]

- sonardepth files can be in one of four ASCII table formats

- 1: format is <time_d sonardepth>
- 2: format is <yr mon day hour min sec sonardepth>
- 3: format is <yr jday hour min sec sonardepth>
- 4: format is <yr jday daymin sec sonardepth>
- time_d = decimal seconds since 1/1/1970
- daymin = decimal minutes start of day
- sonardepth = sonar depth positive down, meters

DATA CUTTING:

DATAUTCLEAR

removes all existing data cutting commands

DATAUTCUT kind mode min max

adds new data cutting command, where:

kind = 0 : cut applied to bathymetry data

kind = 1 : cut applied to amplitude data

kind = 2 : cut applied to sidescan data

mode = 0 : no data are flagged or zeroed

mode = 1 : min and max indicate start and end
beam/pixel numbers between which data
are flagged or zeroed

mode = 2 : min and max indicate start and end
acrosstrack distance (m) between which
data are flagged or zeroed

mode = 3 : min and max indicate minimum and
platform speed (km/hr) between which
data are flagged or zeroed

BATHCUTNUMBER min max

adds new bathymetry data cutting command where
min and max are the start and end beam numbers
between which data are flagged (note that
flagging bathymetry also flags amplitude data)

BATHCUTDISTANCE min max

adds new bathymetry data cutting command where
min and max are the start and end acrosstrack
distance (m) between which data are flagged
(note that flagging bathymetry also flags
amplitude data)

BATHCUTSPEED min max

adds new bathymetry data cutting command where
all beams are flagged for pings with a ship
or vehicle speed less than min or greater than
max (note that flagging bathymetry also flags
amplitude data)

AMPCUTNUMBER min max

adds new amplitude data cutting command where
min and max are the start and end beam numbers
between which amplitude data are zeroed (note
that zeroing amplitude data has no impact on
bathymetry data)

AMPCUTDISTANCE min max

adds new amplitude data cutting command where
min and max are the start and end acrosstrack
distance (m) between which amplitude data are

zeroed (note that zeroing amplitude data has no impact on bathymetry data)

AMPCUTSPEED min max

adds new amplitude data cutting command where all amplitude values are zeroed for pings with a ship or vehicle speed less than min or greater than max (note that zeroing amplitude data has no impact on bathymetry data)

SSCUTNUMBER min max

adds new sidescan data cutting command where min and max are the start and end pixel numbers between which sidescan data are zeroed (note that zeroing sidescan data has no impact on bathymetry data)

SSCUTDISTANCE min max

adds new sidescan data cutting command where min and max are the start and end across-track distance (m) between which sidescan data are zeroed (note that zeroing sidescan data has no impact on bathymetry data)

SSCUTSPEED min max

adds new sidescan data cutting command where all sidescan values are zeroed for pings with a ship or vehicle speed less than min or greater than max (note that zeroing sidescan data has no impact on bathymetry data)

BATHYMETRY EDITING:

EDITSAVEMODE boolean

turns on reading edit save file (from mbedit) [0]

EDITSAVEFILE filename

sets edit save file path (from mbedit) [none]

BATHYMETRY RECALCULATION:

SVPMODE mode

sets usage of a water sound speed model (sound velocity profile, or SVP) [0]

0: bathymetry recalculation by raytracing off

1: bathymetry recalculation by raytracing on

2: translate depths from corrected to uncorrected or vice versa depending on SOUNDSPEEDREF command

SVPMODE filename

sets SVP file path [no default]

SSVMODE boolean

sets surface sound velocity (SSV) mode [0]

0: use SSV from file

1: offset SSV from file (set by SSV command)

2: use constant SSV (set by SSV command)

SSV constant/offset

sets SSV value or offset (m/s) [1500.0]

ANGLEMODE mode

sets handling of beam angles during

raytracing [1]

0: angles not changed before raytracing

1: angles adjusted using Snell's Law for the difference between the surface sound velocity (SSV) and the sound speed at the sonar depth in the SVP.

2: angles adjusted using Snell's Law and the sonar array geometry for the difference between the surface sound velocity (SSV) and the sound speed at the sonar depth in the SVP.

TTMULTIPLY multiplier

sets value multiplied by travel times [1.0]

SOUNDSPEEDREF boolean

determines the handling of the sound speed reference for bathymetry [1]

– note: if raytracing is turned off then this command implies correcting or uncorrecting using the SVP specified with the SVPFILE command

0: produce "uncorrected" bathymetry referenced to a uniform 1500 m/s water sound speed model.

1: produce "corrected" bathymetry referenced to a realistic water sound speed model.

STATIC BEAM BATHYMETRY OFFSETS:

STATICMODE mode

sets offsetting of bathymetry by per-beam statics [0]

0: static correction off

1: static correction by beam number

2: static correction by acrosstrack beam angle

STATICFILE filename

sets static per-beam file path [no default]

– static files are two-column ascii tables

– if correction is by beam number then the beam # is in column 1 and

the depth offset is in m in column 2

– if correction is by beam angle then the beam angle (starboard positive)

is in column 1 and

the depth offset is in m in column 2

DRAFT CORRECTION:

DRAFTMODE mode

sets draft correction [0]

– note: draft merged from navigation before draft correction applied

0: no draft correction

1: draft correction by offset

2: draft correction by multiply

3: draft correction by offset and multiply

4: draft set to constant

DRAFT constant

sets draft value (m) [0.0]

DRAFTOFFSET offset

sets value added to draft (m) [0.0]

DRAFTMULTIPLY multiplier

sets value multiplied by draft [1.0]

HEAVE CORRECTION:

HEAVEMODE mode

sets heave correction [0]

– note: heave correction by offset and/or multiplication is added to any lever heave correction, and then either used in bathymetry recalculation or added to existing bathymetry

0: no heave correction

1: heave correction by offset

2: heave correction by multiply

3: heave correction by offset and multiply

HEAVEOFFSET offset

sets value added to heave (m)

HEAVEMULTIPLY multiplier

sets value multiplied by heave

LEVER CORRECTION:

LEVERMODE mode

sets heave correction by lever calculation [0]

– note: lever heave correction is added to any heave correction by offset and/or multiplication, and then either used in bathymetry recalculation or added to existing bathymetry

0: no lever calculation

1: heave correction by lever calculation

VRUOFFSETX constant

sets athwartships offset of attitude sensor (m)

– note: positive to starboard

VRUOFFSETY constant

sets fore-aft offset of attitude sensor (m)

– note: positive forward

VRUOFFSETZ constant

sets vertical offset of attitude sensor (m)

– note: positive down

SONAROFFSETX constant

sets athwartships offset of sonar receive array (m)

– note: positive to starboard

SONAROFFSETY constant

sets fore-aft offset of sonar receive array (m)

– note: positive forward

SONAROFFSETZ constant

sets vertical offset of sonar receive array (m)

– note: positive down

ROLL CORRECTION:

ROLLBIASMODE mode

sets roll correction [0]

0: no roll correction

1: roll correction by single roll bias

2: roll correction by separate port and
starboard roll bias

ROLLBIAS offset

sets roll bias (degrees)

ROLLBIASPORT offset

sets port roll bias (degrees)

ROLLBIASSTBD offset

sets starboard roll bias (degrees)

PITCH CORRECTION:

PITCHBIASMODE mode

sets pitch correction [0]

0: no pitch correction

1: pitch correction by pitch bias

PITCHBIAS offset

sets pitch bias (degrees)

HEADING CORRECTION:

HEADINGMODE mode

sets heading correction [no heading correction]

– note: heading merged from navigation before
heading correction applied

0: no heading correction

1: heading correction using course
made good

2: heading correction by offset

3: heading correction using course
made good and offset

HEADINGOFFSET offset

sets value added to heading (degrees)

TIDE CORRECTION:

TIDEMODE mode

sets tide correction [0]

– note: tide added to bathymetry after
all other calculations and corrections

0: tide correction off

1: tide correction on

TIDEFILE filename

sets tide file path

TIDEFORMAT constan

sets tide file format [1]

– tide files can be in one of four ASCII
table formats

1: format is <time_d tide>

2: format is <yr mon day hour min sec tide>

- 3: format is <yr jday hour min sec tide>
- 4: format is <yr jday daymin sec tide>
- time_d = decimal seconds since 1/1/1970
- daymin = decimal minutes start of day

AMPLITUDE CORRECTION:

AMPCORRMODE boolean

sets correction of amplitude for
amplitude vs grazing angle function

0: amplitude correction off

1: amplitude correction on

AMPCORRFILE filename

sets amplitude correction file path
[no default]

AMPCORRTYPE mode

sets sidescan correction type [0]

0: correction by subtraction (dB scale)

1: correction by division (linear scale)

AMPCORRSYMMETRY boolean

forces correction function to be symmetric [1]

AMPCORRANGLE constant

sets amplitude correction reference angle
(deg) [30.0]

AMPCORRSLOPE mode

sets amplitude correction slope mode [0]

0: local slope ignored in calculating correction

1: local slope used in calculating correction

2: topography grid used in calculating correction
but slope ignored

3: local slope from topography grid used in
calculating correction

SIDESCAN CORRECTION:

SSCORRMODE boolean

sets correction of sidescan for
amplitude vs grazing angle function

0: sidescan correction off

1: sidescan correction on

SSCORRFILE filename

sets sidescan correction file path
[no default]

SSCORRTYPE mode

sets sidescan correction type [0]

0: correction by subtraction (dB scale)

1: correction by division (linear scale)

SSCORRSYMMETRY boolean

forces correction function to be symmetric [1]

SSCORRANGLE constant

sets sidescan correction reference angle
(deg) [30.0]

SSCORRSLOPE mode

sets sidescan correction slope mode [0]

0: local slope ignored in calculating correction

- 1: local slope used in calculating correction
- 2: topography grid used in calculating correction but slope ignored
- 3: local slope from topography grid used in calculating correction

AMPSSCORRTPOFILE

Sets topography grid used for correcting amplitude and sidescan

SIDECAN RECALCULATION:

SSRECALCMODE boolean

sets recalculation of sidescan for Simrad multibeam data

0: sidescan recalculation off

1: sidescan recalculation on

SSPIXELSIZE constant

sets recalculated sidescan pixel size (m) [0.0]

– a zero value causes the pixel size to be recalculated for every data record

SSSWATHWIDTH constant

sets sidescan swath width (degrees) [0.0]

– a zero value causes the swath width to be recalculated for every data record

SSINTERPOLATE constant

sets sidescan interpolation distance (number of pixels)

METADATA INSERTION:

METAVESSEL string

sets mbinfo metadata string for vessel

METAINSTITUTION string

sets mbinfo metadata string for vessel operator institution or company

METAPLATFORM string

sets mbinfo metadata string for sonar platform (ship or vehicle)

METASONAR string

sets mbinfo metadata string for sonar model name

METASONARVERSION string

sets mbinfo metadata string for sonar version (usually software version)

METACRUISEID string

sets mbinfo metadata string for institutional cruise id

METACRUISENAME string

sets mbinfo metadata string for descriptive cruise name

METAPI string

sets mbinfo metadata string for principal investigator

METAPIINSTITUTION string

sets mbinfo metadata string for principal

investigator
METACLIENT string
sets mbinfo metadata string for data owner
(usually PI institution)
METASVCORRECTED boolean
sets mbinfo metadata boolean for sound
velocity corrected depths
METATIDECORRECTED boolean
sets mbinfo metadata boolean for tide
corrected bathymetry
METABATHEDITMANUAL boolean
sets mbinfo metadata boolean for manually
edited bathymetry
METABATHEDITAUTO boolean
sets mbinfo metadata boolean for automatically
edited bathymetry
METAROLLBIAS constant
sets mbinfo metadata constant for roll bias
(degrees + to starboard)
METAPITCHBIAS constant
sets mbinfo metadata constant for pitch bias
(degrees + forward)
METAHEADINGBIAS constant
sets mbinfo metadata constant for heading bias
METADRAFT constant
sets mbinfo metadata constant for vessel draft (m)

PROCESSING KLUGES:

KLUGE001 boolean
enables correction of travel times in
Hydrosweep DS2 data from the R/V Maurice
Ewing in 2001 and 2002.
KLUGE002 boolean
enables correction of draft values in
Simrad data
– some Simrad multibeam data has had an
error in which the heave has been added
to the sonar depth (draft for hull
mounted sonars)
– this correction subtracts the heave
value from the sonar depth
KLUGE003 boolean
enables correction of beam angles in
SeaBeam 2112 data
– a data sample from the SeaBeam 2112 on
the USCG Icebreaker Healy (collected on
23 July 2003) was found to have an error
in which the beam angles had 0.25 times
the roll added
– this correction subtracts $0.25 * \text{roll}$
from the beam angles before the bathymetry
is recalculated by raytracing through a
water sound velocity profile

- the mbprocess parameter files must be set to enable bathymetry recalculation by raytracing in order to apply this correction

KLUGE004 boolean

deletes survey data associated with duplicate or reversed time tags

- if survey data records are encountered with time tags less than or equal to the last good time tag, an error is set and the data record is not output to the processed data file.

KLUGE005 boolean

replaces survey record timestamps with timestamps of corresponding merged navigation records

- this feature allows users to fix timestamp errors using MBnavedit and then insert the corrected timestamps into processed data

KLUGE006 boolean

changes sonar depth / draft values without changing bathymetry values

KLUGE007 boolean

processing kluge 007 (not yet defined)

- occasionally odd processing problems will occur that are specific to a particular survey or sonar version
- mbprocess will allow one-time fixes to be defined as "kluges" that can be turned on through the parameter files.

ANCILLARY DATA FILES

MB-System also uses a number of ancillary data files, most of which relate to **mbprocess** in some way. By default, these ancillary data files are named by adding a short suffix to the primary data file name (e.g. ".par", ".svp", ".esf", ".nve")

The common ancillary files are listed below. The example names given here follow from an input swath data file name of mydata.mb71.

The processing parameter file used by **mbprocess** has an ".par" suffix. These files are generated or modified by **mbset**, **mbedit**, **mbnavedit**, **mbvelocitytool**, **mbnavadjust**, and **mbclean**.

mydata.mb71.par

The most prominent ancillary files are metadata or "inf" files (created from the output of **mbinfo**). Programs such as **mbgrid** and **mbm_plot** try to check "inf" files to see if the corresponding data files include data within desired areas. The program **mbprocess** automatically generates an "inf" file for any processed output swath file. Also, the program **mbdatalist** is often used to create or update "inf" files for large groups of swath data files.

mydata.mb71.inf

mydata.mb71.inf

The "fast bath" or "fbt" files are generated by copying the swath bathymetry to a sparse, quickly read format

(format 71). Programs such as **mbgrid**, **mbswath**, and **mbcontour** will try to read "fbt" files instead of the full data files whenever only bathymetry information are required. The program **mbprocess** automatically generates an "fbt" file for any processed output swath file. Also, the program **mbdatalist** is often used to create or update "fbt" files for large groups of swath data files. These files are not generated or used when the original swath data is already in a compact bathymetry-only data format.

mydata.mb71.fbt

The "fast nav" or "fnv" files are just ASCII lists of navigation generated using **mblist** with a **-OtMXYHSc** option. Programs such as **mbgrid**, **mbswath**, and **mbcontour** will try to read "fnv" files instead of the full data files whenever only navigation information are required. These files are not generated or used when the original data is already in a single-beam or navigation data format.

mydata.mb71.fnv

The bathymetry edit save file generated by **mbedit** and **mbclean** has an ".esf" suffix.

mydata.mb71.esf

A water sound velocity profile (SVP) file generated by **mbvelocitytool** has an ".svp" suffix unless the user specifies otherwise.

mydata.mb71.svp

Water sound velocity profile (SVP) files generated by **mbsvplist** also use the ".svp" suffix. However, multiple SVP files may be extracted from each input swath file, so the files are numbered using a "_YYY.svp" suffix, where YYY increments from 001.

mydata.mb71_001.svp

mydata.mb71_002.svp

mydata.mb71_003.svp

Edited navigation files generated by **mbnavedit** have an ".nve" suffix:

mydata.mb71.nve

These navigation files can be read independently using format 166.

Adjusted navigation files generated by **mbnavadjust** have an ".naY" suffix, where "Y" is a number between 0-9. The **mbnavadjust** package may be used multiple times for a survey; the adjustments are numbered sequentially from "0":

mydata.mb71.na0

mydata.mb71.na1

mydata.mb71.na2

and so on. These navigation files can be read independently using format 166.

MB-SYSTEM AUTHORSHIP

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OPTIONS

- E** This option causes **mbset** to require all processing modes to be explicitly set by **-P** calls. Normally, **mbset** will implicitly set some modes based on the parameters applied. For example, if a user applies **-PSVPFILE:best.svp**, **mbset** would normally set the bathymetry recalculation mode on so that the SVP file specified actually gets used.
- F** *format*
Sets the data format id used to read the swath data file(s) specified with the **-I** option, overriding the default automatic format detection based on the *infile* name (or the ".mbio_defaults" file). See the **MBIO** manual page for a list of the supported data formats and their id numbers.
- I** *infile*
Swath data file for which an **mbprocess** parameter file will be created or updated, or a datalist file containing a list of input swath data files and/or other datalist files. If *infile* is a datalist file, then **mbset** will attempt to update or create the parameter files for all swath data files identified by recursively reading *infile*. Default *infile*: datalist.mb-1
- L**
This option causes **mbset** to look for any edit save file and navigation file with expected names (based on the input file name) and, if found, set the **mbprocess** parameters so that these files are used. This is a simple on/off flag; it has no additional effect if given more than once. By default (i.e. without **-L**), **mbset** does not search for these ancillary files and only uses parameter values already present in the existing **mbprocess** parameter file or explicitly set with the **-P** option.
- N**
This option causes **mbset** to remove any reference to an adjusted navigation file generated by **mbnavadjust**. If an **mbnavadjust** navigation adjustment file (NAVADJFILE) is referenced in the **mbprocess** parameter file, that file is deleted and the NAVADJFILE and NAVADJMODE parameters are reset to their off/empty defaults.
- H** This "help" flag causes the program to print out a description of its operation and then exit immediately.
- P** *PARAMETER:value* The **-P** option of **mbset** is used to modify a single **mbprocess** parameter command. Here *PARAMETER* may be any of the processing parameter names listed above, and *value* is the corresponding value to be set. This option can be invoked as many times as desired on the command line, allowing **mbset** to set multiple **mbprocess** processing parameters and modes. The separator between *PARAMETER* and *value* can be either ':' or '='. Only the first ':' or '=' acts as a separator; later instances of these characters are taken as part of *value*.
- V** Normally, **mbset** works "silently" without outputting anything to the stderr stream. If the **-V** flag is given, then **mbset** works in a "verbose" mode and outputs the program version being used and lists the processing parameters output to the **mbprocess** parameter file.

EXAMPLES

Suppose the user has a Simrad EM120 data file called "0051_20010829_223755.mb57" that requires processing.

Editing the bathymetry data in this file with **mbedit** will generate an edit save file "0051_20010829_223755.mb57.esf" and an **mbprocess** parameter file "0051_20010829_223755.mb57.par". The contents of the parameter file are:

```
## MB-System processing parameter file
## Written by mb_pr_writepar version $Id$
## MB-system Version 5.0.beta22
## Generated by user <caress> on cpu <menard> at <Fri Sep 6 21:27:41 2002>
##
```

```
##
## Forces explicit reading of parameter modes.
EXPLICIT
##
## General Parameters:
FORMAT 57
INFILE /data/0051_20010829_223755.mb57
OUTFILE /data/0051_20010829_223755p.mb57
##
## Navigation Merging:
NAVMODE 0
NAVFILE /data/0051_20010829_223755.mb57.nve
NAVFORMAT 0
NAVHEADING 0
NAVSPEED 0
NAVDRAFT 0
NAVATTITUDE 0
NAVINTERP 0
NAVTIMESHIFT 0.000000
##
## Navigation Offsets and Shifts:
NAVSHIFT 0
NAVOFFSETX 0.000000
NAVOFFSETY 0.000000
NAVOFFSETZ 0.000000
NAVSHIFTLON 0.000000
NAVSHIFTLAT 0.000000
##
## Adjusted Navigation Merging:
NAVADJMODE 0
NAVADJFILE
NAVADJINTERP 0
##
## Attitude Merging:
ATTITUDEMODE 0
ATTITUDEFILE
ATTITUDEFORMAT 1
##
## Sonardepth Merging:
SONARDEPTHMODE 0
SONARDEPTHFILE
SONARDEPTHFORMAT 1
##
## Data cutting:
DATACUTCLEAR
##
## Bathymetry Flagging:
EDITSAVEMODE 1
EDITSAVEFILE /data/0051_20010829_223755.mb57.esf
##
## Bathymetry Recalculation:
SVPMODE 0
SVPFILE
```



```
SSVMODE 0
SSV 0.000000
TTMODE 0
TTMULTIPLY 1.000000
ANGLEMODE 0
SOUNDSPEEDREF 1
##
## Draft Correction:
DRAFTMODE 0
DRAFT 0.000000
DRAFTOFFSET 0.000000
DRAFTMULTIPLY 1.000000
##
## Heave Correction:
HEAVEMODE 0
HEAVEOFFSET 0.000000
HEAVEMULTIPLY 1.000000
##
## Lever Correction:
LEVERMODE 0
VRUOFFSETX 0.000000
VRUOFFSETY 0.000000
VRUOFFSETZ 0.000000
SONAROFFSETX 0.000000
SONAROFFSETY 0.000000
SONAROFFSETZ 0.000000
##
## Roll Correction:
ROLLBIASMODE 0
ROLLBIAS 0.000000
ROLLBIASPORT 0.000000
ROLLBIASSTBD 0.000000
##
## Pitch Correction:
PITCHBIASMODE 0
PITCHBIAS 0.000000
##
## Heading Correction:
HEADINGMODE 0
HEADINGOFFSET 0.000000
##
## Tide Correction:
TIDEMODE 0
TIDEFILE
TIDEFORMAT 1
##
## Amplitude Correction:
AMPCORRMODE 0
AMPCORRFILE
AMPCORRTYPE 0
AMPCORRSYMMETRY 1
AMPCORRANGLE 30.000000
AMPCORRSLOPE 0
```

```

##
## Sidescan Correction:
SSCORRMODE 0
SSCORRFILE
SSCORRTYPE 0
SSCORRSYMMETRY 1
SSCORRANGLE 30.000000
SSCORRSLOPE 0
##
## Sidescan Recalculation:
SSRECALCMODE 0
SSPIXELSIZE 0.000000
SSSWATHWIDTH 0.000000
SSINTERPOLATE 0
##
## Metadata Insertion:
METAVESSEL
METAINSTITUTION
METAPLATFORM
METASONAR
METASONARVERSION
METACRUISEID
METACRUISENAME
METAPI
METAPIINSTITUTION
METACLIENT
METASVCORRECTED -1
METATIDECORRECTED -1
METABATHEDITMANUAL -1
METABATHEDITAUTO -1
METAROLLBIAS 0.000000
METAPITCHBIAS 0.000000
METAHEADINGBIAS 0.000000
METADRAFT 0.000000
##
## Processing Kluges:

```

Editing the navigation with `mbnavedit` will generate a navigation file named "0051_20010829_223755.mb57.nve" and will modify the parameter file. The changed lines in "0051_20010829_223755.mb57.par" are:

```

## Navigation Merging:
NAVMODE 1
NAVFILE /data/0051_20010829_223755.mb57.nve
NAVFORMAT 9
NAVHEADING 1
NAVSPEED 1
NAVDRAFT 1
NAVATTITUDE 1

```

At this point, running **mbprocess** on "0051_20010829_223755.mb57" will apply the bathymetry flags from **mbedit** and merge the navigation from **mbnavedit**, but will not modify the data in any other way.

If the user wants to recalculate the bathymetry using an SVP file "0051_20010829_223755.mb57.svp" and a roll bias correction of +0.5 degrees, the following will suffice:

```
mbset -I 0051_20010829_223755.mb57 -PSVPFILE:0051_20010829_223755.mb57.svp
-PROLLBIAS:0.5 -PDRAFT:1.95 -V
```

The affected lines in "0051_20010829_223755.mb57.par" are:

```
##
## Bathymetry Recalculation:
SVPMODE 1
SVPFILE 0051_20010829_223755.mb57.svp
SSVMODE 0
SSV 0.000000
TTMODE 0
TTMULTIPLY 1.000000
ANGLEMODE 0
SOUNDSPEEDREF 1
##
## Draft Correction:
DRAFTMODE 4
DRAFT 1.950000
DRAFTOFFSET 0.000000
DRAFTMULTIPLY 1.000000
##
## Roll Correction:
ROLLBIASMODE 1
ROLLBIAS 0.500000
ROLLBIASPORT 0.000000
ROLLBIASSTBD 0.000000
```

To process the data, run mbprocess:

```
mbprocess -I0051_20010829_223755.mb57 -V
```

The output to the terminal is:

```
Program mbprocess
MB-System Version 5.0.beta07
```

```
Program <mbprocess>
MB-system Version 5.0.beta07
```

```
Program Operation:
Input file: 0051_20010829_223755.mb57
Format: 57
Files processed only if out of date.
Comments embedded in output.
```

```
Data processed – out of date:
Input: 0051_20010829_223755.mb57
Output: 0051_20010829_223755p.mb57
```

Input and Output Files:

Format: 57
Input file: 0051_20010829_223755.mb57
Output file: 0051_20010829_223755p.mb57
Comments in output: ON

Navigation Merging:

Navigation merged from navigation file.
Heading merged from navigation file.
Speed merged from navigation file.
Draft merged from navigation file.
Navigation file: 0051_20010829_223755.mb57.nve
Navigation algorithm: linear interpolation
Navigation time shift: 0.000000

Navigation Offsets and Shifts:

Navigation positions not shifted.

Adjusted Navigation Merging:

Navigation not merged from adjusted navigation file.
Adjusted navigation file:
Adjusted navigation algorithm: linear interpolation

Data Cutting:

Data cutting disabled.

Bathymetry Editing:

Bathymetry edits applied from file.
Bathymetry edit file: 0051_20010829_223755.mb57.esf

Bathymetry Recalculation:

Bathymetry recalculated by raytracing.
SVP file: 0051_20010829_223755.mb57.svp
SSV not modified.
SSV offset/constant: 0.000000 m/s
Travel time multiplier: 1.000000 m

Bathymetry Water Sound Speed Reference:

Output bathymetry reference: CORRECTED
Depths recalculated as corrected

Draft Correction:

Draft set to constant.
Draft constant: 1.950000 m
Draft offset: 0.000000 m
Draft multiplier: 1.000000 m

Heave Correction:

Heave not modified.
Heave offset: 0.000000 m
Heave multiplier: 1.000000 m

Lever Correction:

Lever calculation off.

Tide Correction:

Tide calculation off.

Roll Correction:

Roll offset by bias.

Roll bias: 0.500000 deg

Port roll bias: 0.000000 deg

Starboard roll bias: 0.000000 deg

Pitch Correction:

Pitch not modified.

Pitch bias: 0.000000 deg

Heading Correction:

Heading not modified.

Heading offset: 0.000000 deg

Amplitude Corrections:

Amplitude correction off.

Sidescan Corrections:

Sidescan correction off.

Sidescan Recalculation:

Sidescan not recalculated.

Sidescan pixel size: 0.000000

Sidescan swath width: 0.000000

Sidescan interpolation: 0

Metadata Insertion:

Metadata vessel:

Metadata institution:

Metadata platform:

Metadata sonar:

Metadata sonarversion:

Metadata cruiseid:

Metadata cruisename:

Metadata pi:

Metadata piinstitution:

Metadata client:

Metadata svcorrected: -1

Metadata tidecorrected -1

Metadata batheditmanual -1

Metadata batheditauto: -1

Metadata rollbias: 0.000000

Metadata pitchbias: 0.000000

Metadata headingbias: 0.000000

Metadata draft: 0.000000

236 navigation records read

Nav start time: 2001 08 29 22:38:02.082999

Nav end time: 2001 08 29 23:37:22.322000

47 bathymetry edits read

236 input data records

3587 input nav records

17 input comment records

6617 input other records

236 output data records

3587 output nav records

64 output comment records

6617 output other records

Generating inf file for 0051_20010829_223755p.mb57

SEE ALSO

mbsystem(1), mbprocess(1), mbedit(1), mbnavedit(1), mbvelocitytool(1)

BUGS

Oh yeah...